

LLM audit-logic hop

Detecting a sudden change in the reviewer map $P(Y | X)$

OnlineRFPerm landing note (RFPerm-as-Judge)

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Abstract

The LLM **audit** / **moderation** / **preference-judge** stream is the alignment counterpart of hallucination regime detection. Here Y is the reviewer decision, not a factuality bit. The **objective** is whether the audit map hopped: RFPerm / PO as a distributional distance, a permutation test of predictive degradation. The consecutive error *ratio* is only an engineering gate. Domain AUC and `po_risk0` AUROC are sidecars—they score separability, which is not that objective. Quiet vs. fire is the reading. HH-RLHF (Bai et al., 2022): quiet $t = 2$ silent, cut $t = 4$ fires with delay 0.

1 What is being tested

Each batch is (X_t, Y_t) : content to review, and the current auditor’s label. The previous-window probe μ_0 is yesterday’s audit logic. Fire is a yes/no hop of that map. A ratio threshold γ is how the trigger is implemented, not the quantity we claim. High reject rate is not a hop. A stable-strict auditor is still smooth. AUC of a domain discriminator does not match this objective.

Table 1: Objective versus instrumentation. The method asks whether the audit map hopped. Ratio is a trigger. AUC scores a different problem.

Object	Role	Matches the objective?
Last-two fire / RFPerm vs launch probe	did $P(Y X)$ hop / drift	yes
$e_{\text{now}}/e_{\text{prev}}$	engineering gate only	no
Domain AUC / <code>po_risk0</code> AUROC	separability sidecar	no
Reject rate / pos-rate	level of Y , not a map hop	no

Table 2: Two layers for LLM audit. Smoothness is consecutive-window changepoint; trustworthiness is degradation versus a frozen launch probe. A slowly stricter policy can stay smooth; an overnight policy/judge swap is the hop.

Question	Statistic	Audit reading
Is the audit chain still smooth?	last-two hop (fire yes/no)	same rubric, same judge: do adjacent batches still look like one map?
Is the auditor still trustworthy?	RFPerm vs frozen launch probe	even a silky tightening can be caught before the old gold label is unusable.

2 Two layers

Smoothness (last-two): did the rubric break between $t - 1$ and t ? **Trustworthiness** (frozen launch probe): even a silky week-by-week tightening can be rejected versus the original gold before

Table 3: RFPerm-as-Judge on Anthropic HH-RLHF helpful-base ($n = 4000$, 50 batches $\times$ 80, cut B_4). Read fire, not the ratio. Quiet $t = 2$ silent; cut hops in-window (delay 0).

Window	Fire	Reading
Quiet $t = 2$	no	audit map still smooth
Cut $t = 4$	yes	preference map hopped
Post-cut	—	new audit law, then can be smooth again

Table 4: Sidecars on the HH-RLHF cut (not the objective). Style AUC says a $P(X)$ register also moved; text AUC near $1/2$ says this was not a bag-of-words topic swap. Neither is the hop test.

Sidecar	What it is allowed to say
Style-domain AUC	register portrait moved; refresh routing features
Text-domain AUC	lexical mix did not drive the call
Gate ratio / err ratio	log of the trigger, not a score we optimize

the auditor is unusable. Keeping only last-two would miss the boiling-frog. Keeping only the launch probe would call every slow tightening a crash. The robust serving rule uses both.

3 What fire is allowed to mean

On fire: the current judge is not gold; open a human contrast window on Top- k `po_risk0`; cap or refuse DPO/RLHF merges from that window. On quiet: standard path, $w = 1$. `po_risk0` ranks rows after a hop. It is not the objective, not a violation score, not AUC.

References

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, Scott Johnston, Shauna Kravec, Liane Lovitt, Neel Nanda, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, Ben Mann, and Jared Kaplan. Training a helpful and harmless assistant with reinforcement learning from human feedback, 2022. URL <https://arxiv.org/abs/2204.05862>.

Table 5: Serving gate. Fire opens a contrast window; it is not a violation classifier. `po_risk0` only ranks rows after a hop.

State	Meaning	Action
quiet	consecutive audit logic holds	standard path; merge prefs under the old gate; $w = 1$
fire	audit logic hopped	current judge is not gold; human Top- k ; refuse or cap DPO merge; optional $\sqrt{PO}$ on $T = 1$ only